k-means clustering using the Iris dataset

Python codes for k-means Clustering using the Iris dataset

1. Import Libraries

In [1]:
```
import numpy as np
import matplotlib.pyplot as plt

from sklearn.datasets import load_iris
from sklearn.cluster import KMeans
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA
```

2. Load the Iris Dataset

In [2]:
```
iris = load_iris()

X = iris.data
y = iris.target  # only used for visualization (NOT training)

print("Feature names:", iris.feature_names)
print("Target names:", iris.target_names)
```

```
Feature names: ['sepal length (cm)', 'sepal width (cm)', 'petal length (cm)', 'petal
width (cm)']
Target names: ['setosa' 'versicolor' 'virginica']
```

3. Feature Scaling

In [3]:
```
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
```

4. Apply K-Means Clustering

In [4]:
```
kmeans = KMeans(n_clusters=3, random_state=42, n_init=10)
clusters = kmeans.fit_predict(X_scaled)

centroids = kmeans.cluster_centers_
```

```
C:\Users\admin\anaconda3\Lib\site-packages\sklearn\cluster\_kmeans.py:1419: UserWarn
ing: KMeans is known to have a memory leak on Windows with MKL, when there are less
chunks than available threads. You can avoid it by setting the environment variable
OMP_NUM_THREADS=1.
  warnings.warn(
```

5. Reduce Dimensions for Visualization (PCA)

In [5]:
```
pca = PCA(n_components=2)
X_pca = pca.fit_transform(X_scaled)
centroids_pca = pca.transform(centroids)
```

6. Plot Clusters

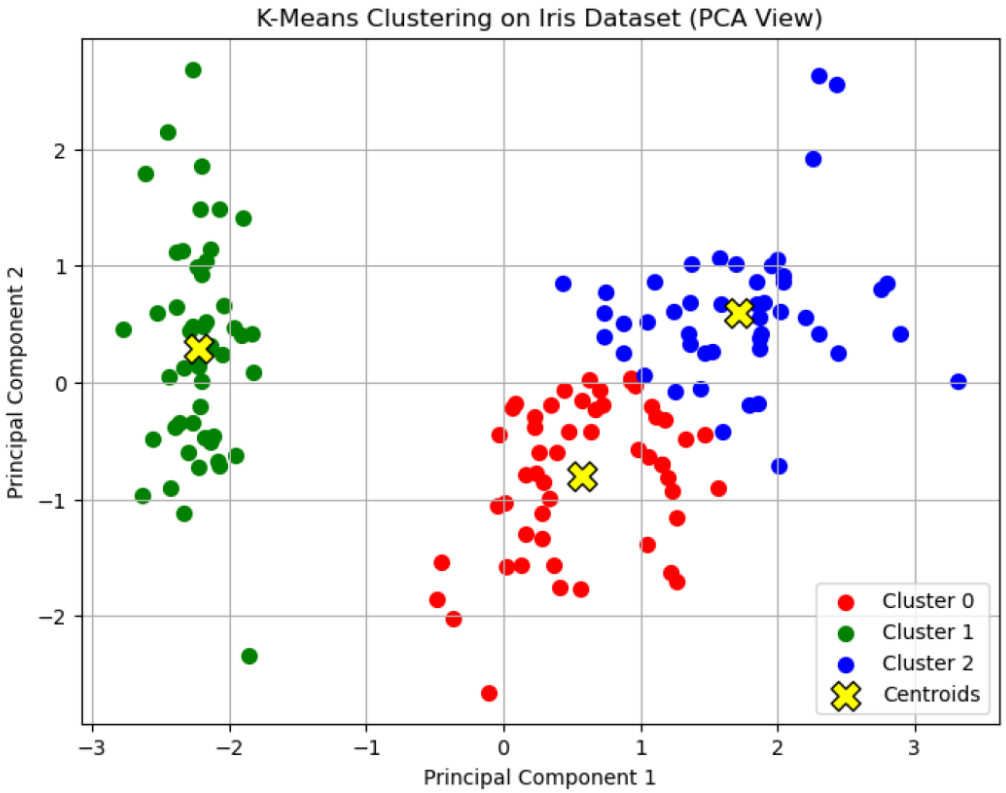
In [6]:
```
plt.figure(figsize=(8, 6))

colors = ['red', 'green', 'blue']

for i in range(3):
    plt.scatter(
        X_pca[clusters == i, 0],
        X_pca[clusters == i, 1],
        s=50,
        color=colors[i],
        label=f"Cluster {i}"
    )

# Plot centroids
plt.scatter(
    centroids_pca[:, 0],
    centroids_pca[:, 1],
    s=200,
    c='yellow',
    marker='X',
    edgecolor='black',
    label='Centroids'
)

plt.title("K-Means Clustering on Iris Dataset (PCA View)")
plt.xlabel("Principal Component 1")
plt.ylabel("Principal Component 2")
plt.legend()
plt.grid()
plt.show()
```


K-Means Clustering on Iris Dataset (PCA View)
Principal Component 2
Principal Component 1
Cluster 0
Cluster 1
Cluster 2
Centroids